

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
clc;
clear;
close all;

%% Filter Specifications
Fs = 1000;    % Sampling Frequency (Hz)
Fp = 100;     % Passband Frequency (Hz)
Fst = 150;    % Stopband Frequency (Hz)
Rp = 1;       % Passband Ripple (dB)
Rs = 40;      % Stopband Attenuation (dB)

%% Normalize Frequencies
Wp = Fp/(Fs/2); % Normalized Passband Frequency
Ws = Fst/(Fs/2); % Normalized Stopband Frequency

%% Calculate Filter Order
[n,Wn] = buttord(Wp,Ws,Rp,Rs);

%% Design Butterworth Low-Pass Filter
[b,a] = butter(n,Wn,'low');

%% Display Filter Details
disp('Butterworth Low-Pass Filter');
disp(['Filter Order = ',num2str(n)]);
disp('Numerator Coefficients (b):');
disp(b);
disp('Denominator Coefficients (a):');
disp(a);

%% Frequency Response
figure;
freqz(b,a,1024,Fs);
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title('Frequency Response of Butterworth Low-Pass Filter');

%% Magnitude and Phase Response
[H,f] = freqz(b,a,1024,Fs);

figure;

subplot(2,1,1);

plot(f,20*log10(abs(H)),'b','LineWidth',2);

grid on;

xlabel('Frequency (Hz)');

ylabel('Magnitude (dB)');

title('Magnitude Response');

subplot(2,1,2);

plot(f,unwrap(angle(H))*180/pi,'r','LineWidth',2);

grid on;

xlabel('Frequency (Hz)');

ylabel('Phase (Degrees)');

title('Phase Response');

%% Impulse Response

figure;

impz(b,a);

grid on;

title('Impulse Response');

%% Pole-Zero Plot

figure;

zplane(b,a);

grid on;

title('Pole-Zero Plot');

%% Filter Coefficients

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fprintf('\nNumerator Coefficients (b):\n');
fprintf('%f ',b);
fprintf('\n\nDenominator Coefficients (a):\n');
fprintf('%f ',a);
%% Test Signal
t = 0:1/Fs:1;
% Low Frequency Signal (Passband)
x1 = sin(2*pi*50*t);
% High Frequency Noise (Stopband)
x2 = 0.5*sin(2*pi*250*t);
% Combined Signal
x = x1 + x2;
% Filtered Output
y = filter(b,a,x);
%% Input vs Output
figure;
subplot(2,1,1);
plot(t,x);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Input Signal');
subplot(2,1,2);
plot(t,y,'r');
grid on;
xlabel('Time (s)');
ylabel('Amplitude');

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```
title('Filtered Output');  
  
%% FFT Analysis  
N = length(x);  
faxis = (0:N-1)*(Fs/N);  
X = abs(fft(x));  
Y = abs(fft(y));  
  
figure;  
subplot(2,1,1);  
plot(faxis,X);  
xlim([0 500]);  
grid on;  
xlabel('Frequency (Hz)');  
ylabel('Magnitude');  
title('Spectrum of Input Signal');  
  
subplot(2,1,2);  
plot(faxis,Y,'r');  
xlim([0 500]);  
grid on;  
xlabel('Frequency (Hz)');  
ylabel('Magnitude');  
title('Spectrum of Filtered Signal');  
  
OUTPUTS:
```



